

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1513

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I Pratip Raj P/192521409 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Pratip Raj P

Assessment Tasks

Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

Answer:

The case describes a retail chain using a **cloud-based inventory system** where POS systems and dashboards communicate with centralized cloud services through APIs. Product data, transaction logs, and analytics datasets are stored in cloud storage, while platform services are used for analytics and demand forecasting.

1. Cloud components used

- **Clients:** POS systems and dashboards.
- **Web APIs:** Connect store devices and clients to centralized cloud services.
- **Cloud Storage:** Stores product data, transaction logs, and analytics datasets.
- **Platform Services:** Used for analytics and demand forecasting.

- **Infrastructure:** Provides the required cloud computing resources.
- **Security:** VPNs, firewalls, and API gateways secure access.

2. How the system works

POS systems collect sales and inventory information from stores. This information is sent through **APIs** to centralized cloud services. The data is stored in cloud storage and analyzed using platform services. Demand forecasting helps the retailer predict future product requirements and maintain suitable inventory levels.

3. Benefits

- Centralized management of inventory data.
- Real-time communication between stores and cloud services.
- Scalable cloud storage for large datasets.
- Better demand forecasting and inventory planning.
- Improved operational efficiency.
- Secure access through VPNs, firewalls, and API gateways.

4. Security measures

The system uses **VPNs, firewalls, and API gateways** to provide secure access to cloud resources and protect inventory and transaction data.

5. Conclusion

The Smart Retail Inventory Management System effectively combines **cloud infrastructure, APIs, cloud storage, and platform services** to manage inventory efficiently, analyze sales data, and forecast demand while maintaining secure access.

Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

Answer:

The case describes an enterprise document collaboration tool similar to Google Docs. Users access it through browsers for real-time editing and sharing, while documents are stored in distributed cloud storage with version control. Web APIs synchronize documents across multiple clients and devices.

1. Cloud components used

- **Clients:** Web browsers on computers and other devices.
- **Web APIs:** Synchronize documents across multiple clients and devices.
- **Cloud Storage:** Stores documents in distributed storage.
- **Version Control:** Maintains different versions of documents.
- **Network Infrastructure:** Provides low latency and high availability.
- **Security Services:** Encryption, identity management, and access permissions.

2. How the system works

Users access the application through a web browser and can **edit and share documents in real time**. The documents are stored in distributed cloud storage, while version control keeps track of changes. Web APIs synchronize updates between different users and devices.

3. Benefits

- Real-time document editing and sharing.
- Access from multiple devices.
- Automatic document synchronization.
- Version control prevents loss of previous changes.
- Distributed storage improves availability.
- Global network infrastructure provides low latency.

4. Security measures

The system uses **encryption, identity management, and access permissions** to protect documents and control who can access or modify them.

5. Conclusion

The Cloud-Based Document Collaboration System combines **cloud storage, web APIs, distributed infrastructure, and security services** to provide real-time, reliable, and secure document collaboration across multiple users and devices.

Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform.

Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Answer:

The case describes a startup building a **cloud-based food ordering system** that is accessed through web browsers and mobile clients. Frontend clients communicate with backend services through RESTful Web APIs hosted on a cloud platform.

1. Cloud components used

- **Clients:** Web browsers and mobile applications.
- **RESTful Web APIs:** Connect frontend clients with backend services.
- **Cloud Storage:** Stores menu images, invoices, and customer data.
- **Virtual Machines:** Provide computing infrastructure.
- **Managed Kubernetes:** Provides scalability and reliability.
- **Cloud Platform Services:** Host and manage backend services.

2. How the system works

Users place food orders through a **web or mobile client**. The frontend communicates with backend services using RESTful APIs. Menu images, invoices, and customer information are stored securely in cloud storage. Virtual machines and managed Kubernetes provide the required scalability and reliability.

3. Security measures

The system uses:

- **Authentication tokens** to verify users.
- **HTTPS** to secure communication.
- **Role-Based Access Control (RBAC)** to control access to resources.

4. Benefits

- Accessible through both web and mobile clients.
- REST APIs allow easy communication between frontend and backend.
- Cloud storage securely manages application data.
- Kubernetes provides scalability and reliability.
- Authentication and RBAC improve security.
- Cloud infrastructure supports real-time operations.

5. Conclusion

The Online Food Ordering Platform integrates **clients, RESTful Web APIs, cloud storage, virtual machines, and managed Kubernetes** to provide a scalable, reliable, and secure food-ordering service.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized